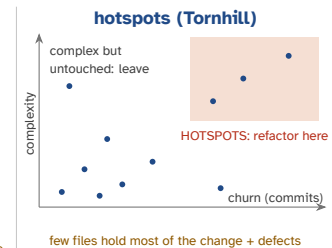
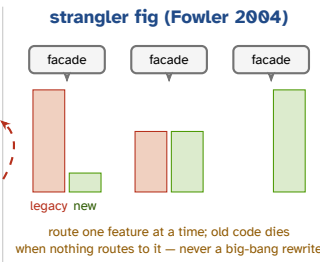
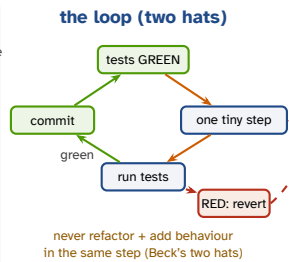
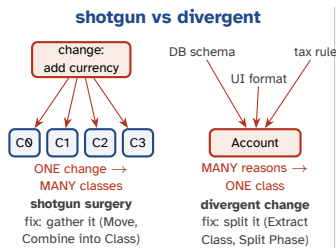


**In one line: Refactoring** (Fowler): “a change made to the internal structure of software to make it easier to understand and cheaper to modify **without changing its observable behaviour**”. A **smell** is a surface symptom that *suggests* a deeper problem — name the smell, name the refactoring, and keep behaviour pinned: **tests first, tiny steps, commit each green step**.



| Smell                    | Tell-tale                                             | Refactoring (Fowler 2nd ed.) · Swift form                                                                 |
|--------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Long Function            | scroll to read; comments label the blocks             | <b>Extract Function</b> , Decompose Conditional, Split Phase                                              |
| Large Class / god object | many fields, many reasons to change; Massive VC       | <b>Extract Class</b> → ViewModel, data source, Coordinator                                                |
| Long Parameter List      | 5+ args; the same group repeats                       | <b>Introduce Parameter Object</b> , Preserve Whole Object; a struct with defaults                         |
| Data Clumps              | lat, lon, accuracy always travel together             | <b>Extract Class</b> / Parameter Object — the clump is a type                                             |
| Primitive Obsession      | String email, Double money, Int status                | <b>Replace Primitive with Object</b> : struct Email validating in init?, Decimal, enum                    |
| Feature Envy             | a method uses another object's data more than its own | <b>Move Function</b> to the data (tell, don't ask)                                                        |
| Divergent Change         | ONE class edited for MANY unrelated reasons           | <b>Extract Class</b> , Split Phase (SRP)                                                                  |
| Shotgun Surgery          | ONE change edits MANY classes                         | <b>Move Function/Field</b> , Combine Functions into Class, Inline Class                                   |
| Repeated Switches        | the same switch type in many places                   | <b>Replace Conditional with Polymorphism</b> — or an enum + exhaustive switch (compiler finds every site) |
| Speculative Generality   | protocol with one conformer, unused hooks “for later” | <b>Collapse Hierarchy</b> , Inline Function/Class, Remove Dead Code                                       |
| Message Chains           | a.b.c.d — the caller knows the whole path             | <b>Hide Delegate</b> (overdone → Middle Man → <b>Remove Middle Man</b> )                                  |
| Temporary Field          | an optional that is nil except in one flow            | <b>Extract Class</b> ; Swift: an enum state with associated values                                        |
| Refused Bequest          | subclass ignores or crashes on what it inherits       | <b>Replace Subclass/Superclass with Delegate</b> (compose), Push Down                                     |
| Comments (deodorant)     | a comment explains <i>what</i> the code does          | <b>Extract Function</b> with that name, Rename                                                            |

**The discipline** No tests → write characterization tests **first** (Feathers: legacy = code without tests): pin *today's* behaviour, bugs included — assert a wrong value, read the actual from the failure, paste it in; snapshot/golden master for big outputs. Then: **one tiny behaviour-preserving step** → run tests → **commit**; red → **revert**, don't debug. **Two hats** (Beck): refactoring *or* adding behaviour, never both in one step/PR. **Preparatory refactoring**: “make the change easy (warning: this may be hard), then make the easy change” (Beck). Prefer IDE refactors (Xcode Rename, Extract to Method) — mechanical = safe. Public interface: **parallel change** (expand → migrate callers → contract), with @available(\*, deprecated, renamed:).

**Big rewrites — strangler fig** Put a facade/router in front; send one feature at a time to the new code; the legacy shrinks until nothing routes to it. iOS: new SwiftUI screens in UIHostingController inside the UIKit app; a protocol in front of the old service with a feature flag choosing the implementation (**branch by abstraction**). Every step ships and can be rolled back. The warning against big-bang: Spolsky, “Things You Should Never Do” (Netscape's rewrite).

```
// BEFORE: primitive obsession + long list + type code
func book(from: String, to: String, date: Date,
         adults: Int, kids: Int, cabin: Int, email: String)
// AFTER: value object, parameter object, enum
struct Email { let raw: String
  init?(s: String) { guard s.contains("@") else { return nil }
    raw = s } }
enum Cabin { case economy, business, first }
struct Trip { var from, to: Airport; var date: Date
  var adults = 1, kids = 0; var cabin = Cabin.economy }
func book(_ trip: Trip, contact: Email)
```

**Remember** Name the smell, name the refactoring · pin behaviour first · one step, green, commit · strangle, don't rewrite · refactor the hotspots.

**When NOT to refactor** Code nobody needs to change (ugly but stable: leave it — refactor where you *work*; boy-scout rule applies to the area you touch) · no tests and no cheap way to get them · right before a release · a published API without a deprecation path · taste alone. A refactoring needs a *payoff*: the next feature, a bug class, onboarding.

**Measure, don't guess Cyclomatic complexity** (McCabe, 1976) =  $E - N + 2P$  on the control-flow graph  $\approx 1 +$  decision points (if, guard, loops, case, &&, ||, ?:, catch); = the minimum number of test paths. McCabe's limit 10; SwiftLint cyclomatic\_complexity warns at 10, errors at 20. **Cognitive complexity** (SonarSource) also penalises nesting. **Hotspots** (Tornhill, *Your Code as a Crime Scene*): change frequency from git log × complexity — refactor the top-right, not the whole codebase. Also: SwiftLint type\_body\_length, function\_parameter\_count; coverage at the change point.

## Traps

- ! **Shotgun surgery** (one change, many classes) ≠ **divergent change** (one class, many reasons) — opposites.
- ! Refactoring without tests is just *changing code*; “refactor” a feature in = two hats.
- ! In Swift a switch over an enum is not the smell — a switch over a *type code* (Int, String) repeated everywhere is.
- ! “Moved the VC's code into a Manager” relocated the smell (god object), not fixed it.